

Plan operations in multi-step problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain? Copy or complete the first step.

2. A club collects £18 from each of 236 members, then spends £1,875. How much remains?

3. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

4. Miro says: Miro begins calculating every number in the order it appears. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain? Copy or complete the first step.
Answer 48 x 125, then subtract 3,760. The answer is 2,240.
2. A club collects £18 from each of 236 members, then spends £1,875. How much remains?
Answer £2,373.
3. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.
Answer Divide 9,450 by 35, then add 12. Each group has 282 items.
4. Miro says: Miro begins calculating every number in the order it appears. Is this correct? Explain.
Answer Plan the relationships first. The order of numbers in the text may not match the calculation order.

Plan operations in multi-step problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A club collects £18 from each of 236 members, then spends £1,875. How much remains?

2. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

3. Write a two-step problem represented by $(64 \times 28) - 950$.

4. Identify and correct this misconception: Miro begins calculating every number in the order it appears.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. A club collects £18 from each of 236 members, then spends £1,875. How much remains?
Answer £2,373.
2. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.
Answer Divide 9,450 by 35, then add 12. Each group has 282 items.
3. Write a two-step problem represented by $(64 \times 28) - 950$.
Answer Answers vary. The final answer to the expression is 842.
4. Identify and correct this misconception: Miro begins calculating every number in the order it appears.
Answer Plan the relationships first. The order of numbers in the text may not match the calculation order.
5. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Plan operations in multi-step problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Write a two-step problem represented by $(64 \times 28) - 950$.

2. Explain why this reasoning fails: Miro begins calculating every number in the order it appears.

3. Create a new example that tests the same idea as: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Write a two-step problem represented by $(64 \times 28) - 950$.

Answer Answers vary. The final answer to the expression is 842.

2. Explain why this reasoning fails: Miro begins calculating every number in the order it appears.

Answer Plan the relationships first. The order of numbers in the text may not match the calculation order.

3. Create a new example that tests the same idea as: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.

Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.

Answer Answers vary. A valid example and counterexample should support the classification.